# GAME THEORY

BY

DOUGLAS QUIGG

RESEARCH AND THEORY
A DOMAIN THAT SUPPLIES ITS OWN ANSWERS, AND THE READING IT CORRECTED

ORIOR
2026

© 2026 Douglas Quigg
AGPL-3.0-or-later OR LicenseRef-Commercial OR LicenseRef-Educational

*For Megan, my loving wife, the only person that really understands what I mean when I speak.*

*For Autumn and Opal, without you this thought would have never occurred.*

*To everyone else, I wish you well.*

## ABSTRACT

The workbook [6] states what this work needs and cannot afford. Every claim about how far the construction reaches is empirical, quantified over a set nobody has enumerated, and getting from a list of instances to a statement about a domain takes a controlled series inside that domain with an outside answer attached to every row. It then prices that at years, because obtaining a row costs a database, a fetch, or a laboratory.

Games remove the price. A subtraction game's Grundy sequence is eventually periodic, proved [1], and its period is a quantity a dull exact routine computes by comparison. A row therefore costs a second of arithmetic, the number of rows is a choice, and no row's answer comes from the instrument being tested. Three hundred and eighty five move sets are generated. Of the three hundred and eighty three whose period is inside the window the detector can score, the detector returned the exact period on three hundred and eighty three, at a margin sixteen to twenty six times the best any shuffle of the same values reached, with no row failing to clear its own floor.

Two older answers sit beside that series. Bouton settled Nim's losing positions in 1901 [2] and every axis line of a Nim grid holds exactly one, on sixty four lines, four thousand and ninety six, and thirty two thousand seven hundred and sixty eight. Wythoff settled his own in 1907 [8], and a grid played out by the minimum excludant recursion, holding no formula, reproduces his closed form exactly and returns the golden ratio to one part in a thousand.

The correction is on the other side of that pair and it reaches past this research paper. Wythoff's positions make a word with no period in it, and passed that word the period detector returns a number anyway. The number clears its own shuffle floor by three to twenty nine times, and no margin test refuses it. It is the denominator of a continued fraction convergent of the word's slope, on sixty one of sixty three rows across nine slopes, and it grows as the window widens: eight, then thirteen, then twenty one, then thirty four, then fifty five. A rotation by an irrational does agree with itself at the denominator of a good rational approximation to it, and the detector reads that agreement correctly. Only the name on the output is false. The diagnostic that separates the two costs nothing and uses no oracle: a real period is the same number at every window, and one hundred and fifteen of one hundred and fifteen periodic games give one answer across five windows where none of nine aperiodic words gives one across seven.

The played half of the research paper reports a negative and retains it. Reversi leaves a departure from the null of twenty one percent, twenty five times the spread between disjoint batches, and a control game on the same board with the flip rule removed sits on the null. So the departure is the flip rule. It is not the play: three arms of measurably different strength, in nine pairings, order themselves clearly by win share and not at all by departure, at a correlation of 0.07.

# CONTENTS


1 WHY A GAME . . . . . 1
1.1 The hole this fills is named in the workbook . . . . . 1
1.2 What a game costs . . . . . 1
1.3 Two types of object, and they are not the same evidence . . . . . 2
1.4 What is not claimed . . . . . 2

2 THE SERIES . . . . . 3
2.1 How a row is made . . . . . 3
2.2 The window limits which rows can be scored . . . . . 3
2.3 Every row against its own floor . . . . . 4
2.4 What the series does and does not license . . . . . 4

3 TWO THEOREMS, AND NEITHER IS THIS WORK'S . . . . . 5
3.1 Bouton, 1901 . . . . . 5
3.2 Wythoff, 1907 . . . . . 5
3.3 Why the pair is here together . . . . . 6

4 WHAT A PERIOD IS NOT . . . . . 7
4.1 The reading . . . . . 7
4.2 What it is actually reading . . . . . 7
4.3 The reading is a capability read backwards . . . . . 8
4.4 The test, and it costs nothing . . . . . 8
4.5 What this does not reach . . . . . 9

5 THE PLAYED BOARD . . . . . 11
5.1 Why a board is worth reading at all . . . . . 11
5.2 The representation, and what happens to an empty square . . . . . 11
5.3 The departure, and the control that says what it is . . . . . 12
5.3.1 The 0.971 row, and the opening it reads . . . . . 12
5.4 The negative . . . . . 12
5.5 Reading the width back off it . . . . . 13

6 WHERE IT STOPS . . . . . 15
6.1 A flattened grid does not carry one period . . . . . 15
6.2 What this research paper has not established . . . . . 16
6.3 What the research paper is for, if one thing . . . . . 16

BIBLIOGRAPHY . . . . . 17

# 1

# WHY A GAME

This chapter says what a game supplies that no other corpus in this work does, and why that is about cost and not about games.

## 1.1 THE HOLE THIS FILLS IS NAMED IN THE WORKBOOK

The workbook [6] divides its own claims in two and the division is not a matter of confidence. One half is proved: a subset of a pattern's points is a necessary condition, the max entropy background under a fixed constraint is unique, a null permutation is distribution-free. Nothing about a corpus bears on any of it, and a hundred agreeing corpora carry the weight of one because the property was in doubt on none of them.

The other half is reach, and the workbook says plainly what closing it would take. A claim covering a whole type of object is empirical, quantified over a set nobody has enumerated, and no run of any length settles it. An eleventh row raises a count and moves the quantifier no distance. What moves it is a controlled series inside one domain, with an outside answer attached to every row, and the workbook prices that at years because obtaining the rows is the expense.

That price is a fact about the domains this work has read and not about the argument. A crystal costs a database and a fetch. A protein costs the same. A corpus of Salishan speech costs papers stored under terms, transcription by hand, and an obligation to the people whose language it is that no amount of arithmetic discharges. In every one of them the reading is cheap and the row is dear.

## 1.2 WHAT A GAME COSTS

A game inverts that. The rules of an impartial game are a few lines, the position space is generated by those lines, and the answer each position carries is a theorem.

Take a subtraction game: a heap of $n$, a finite set $S$ of legal takes, and whoever cannot move loses. The Grundy value of a position is the least non-negative integer absent from the values of the positions it reaches, which is four lines of code, and the sequence of those values is eventually periodic. That last is not an observation. It is proved [1], and the period is a quantity a separate routine finds by sliding the tail against itself and checking equality to the end.

A row of this series runs in five steps: choose a move set, generate the sequence, compute the exact period by comparison, hand the sequence to the detector, compare. The generator and the answer share no code with the detector, the answer is exact and carries no tolerance, and patience

is the only limit on the number of rows. Three hundred and eighty five move sets take under a minute.

The point is not that games are interesting. The point is that this is the first domain in the work where the controlled series the workbook asks for can actually be run, and the workbook's estimate of years is an estimate about corpora that have to be obtained.

## 1.3 TWO TYPES OF OBJECT, AND THEY ARE NOT THE SAME EVIDENCE

This research paper reads both and retains them apart.

An **impartial game** is a position index carrying a value. There is no board in it, no player, and no choice: the value at a position is determined by the rules alone. The series lives here, and the two old theorems sit here with it.

A **played board** is a position: a square is a point and the piece on it is its value. A board carries two ingredients that every other corpus in this work holds welded together. A text carries its grammar and its author at once and there is no way to obtain the same author under a different grammar. A board separates them for nothing, because three players of different strength on one fixed rule set give nine corpora differing in the play alone.

That separation is the played half's whole reason to exist, and what it returns is a negative. The measure reads the rules and not the play. That is recorded in its own chapter with the control that establishes it.

## 1.4 WHAT IS NOT CLAIMED

A game is not a natural object. Its rules were written by a person and its regularity was put there deliberately, and that makes it a positive control and not a discovery. The crystallography research paper [5] makes the same admission about a published cell edge and it applies here with less excuse, since nobody has to go and measure a subtraction game.

A positive control does what a memoryless control cannot. A memoryless corpus returning 1.00 shows an instrument does not invent structure. It cannot show the instrument finds structure that is there. These rows show that, cheaply, at a count that makes a coincidence untenable, and they show something a memoryless control never can: where a reading is confident, clears every floor, and is still not what it is called.

# 2

# THE SERIES

This chapter reports the controlled series and every number in it, including the rows the detector cannot be asked to read.

## 2.1 HOW A ROW IS MADE

A move set is a finite set of positive integers. The game is one heap, a move takes a number in that set, and whoever cannot move loses. Two thousand and forty eight positions are generated, each carrying the least non-negative integer absent from the values it reaches.

The answer is computed twice by two routines that share nothing. The exact one slides the tail of the sequence against itself and returns the shortest shift under which it repeats to the end. The measured one is the period detector, the same one the crystallography research paper [5] asks for a cell edge, which is passed only a list of small integers.

The sweep is every move set of one to four moves drawn from 1 to 10, which is three hundred and eighty five games. Every one of them settled: the exact routine found a period on all three hundred and eighty five, and the periods present run 2 to 45.

## 2.2 THE WINDOW LIMITS WHICH ROWS CAN BE SCORED

The detector scores a candidate period together with its own multiples. The crystallography research paper [5] makes that correction, because the tallest lag reads published cell edges at twice their value on ten of eighteen axes. A period beyond half the window has one multiple inside the window. Its family is then one lag, and the score degenerates to taking the tallest lag again.

Rows whose true period exceeds half the window are therefore out of range by construction. They are counted separately below and they are not failures. Counting them as failures would be reporting the detector's documented precondition as its error rate.

| window | in range | exact | out of range |
|---|---|---|---|
| 16 | 115 | 115 | 270 |
| 24 | 220 | 220 | 165 |
| 32 | 346 | 346 | 39 |
| 48 | 378 | 378 | 7 |
| 64 | 383 | 383 | 2 |

Three hundred and eighty three of three hundred and eighty three at the widest window, and the same at every narrower one over the rows it can reach. The two rows still out of range at a window of 64 have periods of 45 and 33.

## 2.3 EVERY ROW AGAINST ITS OWN FLOOR

A count of agreements is not evidence on its own, because a detector that always answered 2 would agree with every period-2 row. Each sequence is therefore scored against twelve shuffles of itself. A shuffle retains every Grundy value and destroys which position carries it. Whatever margin it reaches, the detector reads off the histogram alone. The figure quoted is the tallest of the twelve instead of their average.

At a window of 64, over three hundred and eighty three rows:

| | minimum | median | max |
|---|---|---|---|
| live margin | 1.016 | 1.091 | 2.000 |
| shuffle floor | 0.030 | 0.044 | 0.066 |

The worst ratio of live margin to its own floor is 16.0 and the median is 25.6. No row fails to clear its floor. The bands do not touch: the smallest live margin is fifteen times the largest floor anywhere in the series.

## 2.4 WHAT THE SERIES DOES AND DOES NOT LICENSE

It licenses a statement about this domain. Over subtraction games with move sets drawn from 1 to 10, the detector returns the exact Grundy period on every sequence whose period it is able to score. The quantifier is over an enumerated set, the answers come from a proof, and the count is large enough that a coincidence would have to be systematic. Luck does not reach three hundred and eighty three.

It does not license a statement about any other domain, and it does not upgrade the workbook's span table. The workbook's own deduction [6] says why: soundness belongs to the construction and cost belongs to the domain's distribution. A result obtained inside one distribution stays inside it. A subtraction game's Grundy sequence is exactly periodic after a short preperiod, with two to a few dozen distinct values and no noise at all. That is the easiest case a period detector can be displayed. The series establishes that the detector is right where an answer exists and is checkable, at a count nobody has to take on faith, and the next chapter is the half that matters more.

# 3

## TWO THEOREMS, AND NEITHER IS THIS WORK'S

This chapter reads two published answers off grids generated without them.

### 3.1 BOUTON, 1901

Nim is heaps, a move takes any number from one heap, and whoever cannot move loses. Bouton [2] showed that a position loses for the player to move exactly when the heap sizes exclusive or to zero.

One consequence of that is readable by counting and needs no exclusive or performed by the reader. Fix every heap but one. Exactly one size of the free heap makes the total zero. Every axis line of the grid therefore holds exactly one losing position.

| game | lines | losses per line | loss density |
|---|---|---|---|
| 2 heaps of 64 | 64 | {1} | 0.01562 |
| 3 heaps of 64 | 4096 | {1} | 0.01562 |
| 4 heaps of 32 | 32768 | {1} | 0.03125 |

The set is one and not a range, on every line, in every arity. A set of the same density scattered over the same cells puts exactly one loss in a line of 64 with probability 0.371. A scatter producing all ones over four thousand and ninety six lines does not happen, and there is no need to quantify it further.

This grid is generated from Bouton's own theorem, which the generator states. These rows are therefore reported as a count and not as a recovery. What they establish is that the property is in the grid where a reader can reach it, without arithmetic that knows what Nim is.

### 3.2 WYTHOFF, 1907

Wythoff's game is two heaps, where a move takes any number from one heap or the same number from both. Wythoff [8] showed the losing positions are the pairs $(\lfloor n\varphi \rfloor, \lfloor n\varphi^2 \rfloor)$ and their mirrors.

The grid here is played out by the minimum excludant recursion and holds no formula. Comparing it against that closed form checks the grid against an answer it was never given.

| side | positions | matches the 1907 form | $a_n/n$ | relative error |
|---|---|---|---|---|
| 200 | 76 | yes | 1.605263 | $7.89 \times 10^{-3}$ |
| 400 | 152 | yes | 1.611842 | $3.83 \times 10^{-3}$ |
| 800 | 305 | yes | 1.616393 | $1.01 \times 10^{-3}$ |

Every pair agrees with the published one, at every side. The second coordinate minus the first runs 1, 2, 3, 4 and onward with no exception, as the other half of Wythoff's statement requires. And the ratio converges on $\varphi = 1.618034$ from below, reaching one part in a thousand at a side of 800, which is a published irrational arriving out of a game played by a recursion that was told nothing.

### 3.3 WHY THE PAIR IS HERE TOGETHER

They are the same shape of game and their answers are opposite in the way this work cares about.

Nim's losing set is built from the exclusive or, which is a bitwise operation, and its regularity is exact at every scale. Wythoff's sits on a line of slope $\varphi$, and a line of irrational slope passes through no two lattice points in the same relation twice. Nothing on it repeats, even approximately or eventually.

So the pair is a positive and a negative control with published answers on both sides, which no other pair of controls in this work supplies. The crystallography research paper [5] argues that a memoryless control can only show an instrument does not invent structure, and that showing it finds structure that is there needs an answer from outside. Nim supplies that answer for nothing. Wythoff supplies the other one: an object with real structure that a shift detector must not report a period on.

The detector reports one. That is the next chapter and it is the reason this research paper exists.

# 4

# WHAT A PERIOD IS NOT

This chapter states a confident reading that is wrong, says exactly what it reads instead, and gives the test that catches it.

## 4.1 THE READING

Take Wythoff's losing positions in order and write down the gaps between successive first coordinates. Two values appear, 1 and 2, in ratio 1.62069 against $\varphi = 1.61803$, and the word they make is Sturmian [4]. It has no period. That is not a measurement and no longer sequence would change it: it follows from $\varphi$ being irrational.

Hand that word to the period detector and it returns a number.

| window | reported |
|---|---|
| 16 | 8 |
| 24 | 8 |
| 32 | 13 |
| 48 | 21 |
| 64 | 21 |
| 96 | 34 |
| 128 | 55 |

Those are Fibonacci numbers. Every one of the seven rows clears its own shuffle floor, and over the full table of nine slopes the margin runs 3.4 to 29 times that floor. No margin test refuses any of them.

## 4.2 WHAT IT IS ACTUALLY READING

The mechanism is continued fractions and it is not mysterious.

The word is the gap sequence of a rotation by an irrational $\alpha$. Such a word agrees with itself shifted by $q$ whenever $q\alpha$ is close to an integer, and the $q$ for which $q\alpha$ is closest to an integer are exactly the denominators of the convergents of $\alpha$. The Fibonacci numbers are the convergent denominators of $\varphi$. So the detector is finding real self-similarity, at the lags where real self-similarity is, and reporting it correctly. The only thing wrong is the name on the output.

The prediction that follows is falsifiable and is tested here. If the mechanism is continued fractions, the reported number should be a convergent denominator of the slope for any irrational slope and not only for $\varphi$. Nine slopes, seven windows each:

| slope | reported, by window |
|---|---|
| $\varphi$ | 8, 8, 13, 21, 21, 34, 55 |
| $\sqrt{2}$ | 5, 12, 12, 17, 29, 29, 29 |
| $e$ | 7, 7, 7, 7, 32, 39, 39 |
| $\pi$ | 7, 7, 7, 7, 7, 7, 53 |
| $\sqrt{3}$ | 4, 11, 15, 15, 15, 41, 56 |
| $\sqrt{5}$ | 4, 4, 13, 17, 17, 17, 55 |
| $\sqrt{7}$ | 3, 11, 14, 17, 31, 48, 48 |
| $\ln 2 + 1$ | 8, 10, 13, 13, 13, 13, 62 |
| $\sqrt[3]{2} + 1$ | 4, 4, 15, 23, 27, 27, 50 |

Sixty one of the sixty three reported numbers are a convergent or intermediate denominator of their own slope. The intermediate fractions are counted because they are also one-sided best approximations and a detector reading agreement has no reason to prefer the two-sided ones.

The two exceptions are the two weakest rows in the table. Sorted by margin, they sit at rank 0 and rank 8 of 63, and the lowest of all of them is one of the two. Where the mechanism does not account for the answer, the detector is also at its least sure.

$\pi$ is the row that tests the mechanism, since a fit alone would not predict it. Its continued fraction has a 292 in it early. Its convergent denominators therefore jump from 7 to 106, and the detector holds 7 across six consecutive windows before moving at all. A slope whose approximations improve slowly, like $\varphi$, moves at every step. The shape of the movement is the shape of the slope's continued fraction.

## 4.3 THE READING IS A CAPABILITY READ BACKWARDS

Nothing here says the detector is broken. Passed a word built from a rotation, its answers across widening windows enumerate the convergent denominators of the rotation's slope, which is to say it recovers the continued fraction of a number it was never given. That belongs in the same list as an image returning its own width and a protein backbone returning its bond lengths.

What it cannot be allowed to do is return that number under the word *period* to a caller who will read it as one. This work has one instrument reporting periods, the crystallography research paper [5] uses it on cell edges, and a reader who believes a high margin means a period believes something the table above refutes.

## 4.4 THE TEST, AND IT COSTS NOTHING

A true period is a property of the sequence. An approximation is a property of the sequence and the window together. So widen the window and see whether the answer moves.

| | one answer at every window |
|---|---|
| subtraction games with a real period, at 5 windows | 115 of 115 |
| aperiodic words, at 7 windows | 0 of 9 |

The separation is total on the rows tested, it needs neither oracle, null nor second instrument, and it costs one extra call per window. The instrument carries it as a second reading beside the period, which returns a period only where widening the window leaves the answer alone and returns nothing where it does not. The single-window reading is retained, because this is a second question and not a correction to the first. Any reading of a period anywhere in this work that is taken at a single window is, on this evidence, unverified. It is not displayed to be wrong. The crystallography rows are safe because a lattice has a period for the reported number to be.

## 4.5 WHAT THIS DOES NOT REACH

The window sweep separates a period from a rotation. It has not been displayed to separate a period from anything else, and the aperiodic words tested are all Sturmian, which is one family. A sequence that is aperiodic in some other way might well hold its answer across windows and defeat the test. Nine slopes is nine, the same way ten domains is ten, and this chapter is subject to the workbook's own rule [6] that an eleventh instance moves the quantifier no distance.

The claim that holds is narrow and it is enough: a confident margin is not evidence of a period, one worked mechanism explains what the confident margin reads instead, and a test that separates the two on every row tried is free.

# 5

# THE PLAYED BOARD

This chapter separates what the rules put on a board from what the players do, and reports the negative that comes out.

## 5.1 WHY A BOARD IS WORTH READING AT ALL

Every corpus in this work holds two things at once and cannot separate them. A text carries its grammar and its author together, and there is no way to obtain the same author writing under different grammar. The workbook's Salishan rows [6] spend real effort on exactly this and the effort does not fully succeed.

A board separates them for nothing. The rules are fixed and public. The play is a choice. Three players of measurably different strength on one rule set give nine corpora that differ in the second alone, and the rule set's contribution is in all nine.

## 5.2 THE REPRESENTATION, AND WHAT HAPPENS TO AN EMPTY SQUARE

A square is a point and the piece on it is its value. Reversi on sixty four squares, two colors.

Emptiness does not get a symbol. A board part way through a game is mostly empty, and a symbol for emptiness puts the game's clock into the histogram, where a measure reading counts will find it and report how far along the game was as though it were arrangement. Measurements that read values drop the empty squares; the one that reads the grid's shape retains them, because a hole in a row is part of the shape.

The null is this work's null permutation written for a grid: the same pieces on the same squares in a uniformly drawn order. It retains how many of each color are on the board and which squares are occupied, and destroys only which color sits where. Everything a rule imposed is in what it destroys.

The measure is the share of occupied squares whose neighbor one step away carries the same color, averaged over both axes. The dispersion measure this work usually quotes needs four symbols clearing an occurrence floor, and a board has two colors. This measure reads the same quantity at the only lag a board is large enough to carry.

## 5.3 THE DEPARTURE, AND THE CONTROL THAT SAYS WHAT IT IS

Two hundred games with both sides drawing uniformly from the legal moves. Live 0.6555 against a null of 0.5350, a ratio of 1.225. Six disjoint batches of two hundred give 1.2253, 1.2193, 1.2219, 1.2238, 1.2222 and 1.2282, a spread of 0.0089. The departure is about twenty five times the spread between batches.

A departure on its own does not say which ingredient produced it, and a board supplies the control for nothing. Play the same board with the same two colors and the same number of squares filled, where a move places a piece and nothing is turned over. That game has rules and has play and has no flip.

| | live | null | ratio |
|---|---|---|---|
| Reversi, 200 games | 0.6555 | 0.5350 | 1.225 |
| place-only, 400 boards, from Reversi's opening | 0.4763 | 0.4904 | 0.971 |
| place-only, 400 boards, from an empty board | 0.4915 | 0.4923 | 0.998 |

The departure is the flip rule. A board being a board contributes nothing.

### 5.3.1 *The 0.971 row, and the opening it reads*

Retained with Reversi's four fixed center pieces, the control reads 0.971, three floors below the null where the argument wants one. The gap is the opening itself. Those four squares are arranged black over white and white over black. All four neighboring pairs inside them therefore disagree by construction, where a draw would have half of them agree. Four guaranteed disagreements out of a hundred and twelve pairs is 0.018 of the reading and accounts for all of the gap. Starting from an empty board returns 0.998.

The row also bounds the Reversi rows. The same four squares sit under every Reversi row above and push them in the same direction. The ratio of 1.225 is understated by about that much and is never overstated.

## 5.4 THE NEGATIVE

The three arms are the textbook ones. Drawing uniformly from the legal moves knows the rules and has no strategy. Taking the move that turns over the max pieces is the standard weak heuristic and is weak here too. Taking a corner where one is offered, and otherwise turning over the fewest pieces, is the standard improvement: a corner can never be flipped back, and a small flip leaves the opponent fewer replies.

Strength is not asserted. It is played out, and the win share is an independent quantity that is not a reading of the departure.

| black v white | live | ratio | black's share of wins |
|---|---|---|---|
| random v random | 0.6555 | 1.225 | 0.380 |
| random v greedy | 0.6581 | 1.191 | 0.398 |
| random v corner | 0.6722 | 1.224 | 0.320 |
| greedy v random | 0.6543 | 1.208 | 0.630 |
| greedy v greedy | 0.6404 | 1.184 | 0.492 |
| greedy v corner | 0.6735 | 1.198 | 0.425 |
| corner v random | 0.6700 | 1.241 | 0.645 |
| corner v greedy | 0.6881 | 1.194 | 0.672 |
| corner v corner | 0.6723 | 1.255 | 0.532 |

The win shares order the arms cleanly and in the expected direction. Corner beats random 0.645 and beats greedy 0.672; greedy beats random 0.630; each arm against itself sits near a half.

The departures do not order them at all. They run 1.184 to 1.255 with no relation to the column beside them, and the correlation over the nine pairings is 0.067.

The explanation is the same one that makes the flip rule the source of the departure. Every legal move turns over a bracketed run, whoever chose it. A strong player and a weak one leave the same type of trace and differ only in where on the board they put it. The measure reads what was done to the board. It does not read who chose it.

## 5.5 READING THE WIDTH BACK OFF IT

An image read as a byte sequence returns its own width. A board is the same shape of object with a width nobody has to look up.

| arm | reading | period | margin |
|---|---|---|---|
| random | row major | 8 | 0.0917 |
| random | column major | 8 | 0.1291 |
| random | scattered null | 7 | 0.0106 |
| corner | row major | 8 | 0.1209 |
| corner | column major | 8 | 0.1368 |
| corner | scattered null | 10 | 0.0093 |

Eight, across the rows and down the columns, on a weak arm and a strong one, at seven to fifteen times what the same pieces scattered reach. The null returns a wrong number at a margin near zero. That is the correct behavior: it is asked for a period and reports one with no confidence at all.

That result has a boundary and the next chapter is where it sits.

# 6

# WHERE IT STOPS

This chapter states what a flattened grid cannot be asked, and what this research paper does not establish.

## 6.1 A FLATTENED GRID DOES NOT CARRY ONE PERIOD

Reversi returns 8 because nothing repeats along a Reversi row. That is a property of the corpus and not a property of the reader, and a reader passed a grid with structure on both axes has no way to know which of them it answered with.

The demonstration is a two-heap subtraction game. The value of a sum of independent games is the exclusive or of their values. That is Sprague and Grundy's theorem [7, 3] and is not measured here. A square grid of two heaps under one move set is therefore generated for nothing. Such a grid repeats down the column at the width it was arranged at, and along the row at the game's own Grundy period. Three hundred and fifteen of them, at five widths:

| the flattened answer | rows |
|---|---|
| the width | 108 |
| the game's own Grundy period | 96 |
| the width off by one, which is diagonal content | 68 |
| a combination of the two | 43 |

The same grids shuffled return the width on 5 of 100, the chance rate. None of the four groups above is noise. Every one of them is a real repeat in the flattened sequence, and the detector returns whichever family scored higher.

The off-by-one group is the clearest case. Where a grid's content is a function of the sum or the difference of its coordinates, as a checkerboard is, the shortest repeat in the flattened sequence is at the width minus one and not at the width, because the lag that returns the same value steps one row down and one column back. The detector reports $W - 1$ and the detector is right. The ground truth of $W$ is the wrong question.

The rule that comes out is short. Read the axes separately where the caller knows there are two. A single flattened period is ambiguous where they do not, and reporting it as the width is an assumption about the corpus dressed as a measurement.

## 6.2 WHAT THIS RESEARCH PAPER HAS NOT ESTABLISHED

**The series covers subtraction games only.** The workbook's own division [6] says why: soundness belongs to the construction and cost belongs to the domain's distribution. A result obtained inside one distribution stays there. A subtraction game's Grundy sequence is exactly periodic after a short preperiod, over a small alphabet, with no noise. It is the easiest case a period detector can be displayed, and three hundred and eighty three rows of it establish that the detector is right where an answer exists and is checkable. They establish nothing about a corpus that is noisy, or only approximately periodic, or long.

**The window test is displayed against one family of counterexample.** Every aperiodic word tested is Sturmian. A sequence aperiodic in some other way might hold its answer across widening windows and defeat the test entirely, and nothing here rules that out. Nine slopes is nine.

**Two of the sixty three Sturmian rows are unexplained.** The continued fraction mechanism accounts for sixty one. The two it does not are the weakest rows in the table by margin, at rank 0 and rank 8 of 63, which is consistent with them being noise and is not a demonstration that they are.

**The played board is one game.** Reversi has a flip rule, and the finding that the departure reads the rules and not the play may be a fact about flip rules. A game whose rules constrain where a piece is placed, and flip nothing, might well leave a trace that a stronger player deepens. Nothing here has tried one.

**The three arms are weak.** Corner beats random by 0.645 to 0.355 and that is a real ordering, but all three are one line of heuristic. The correlation of 0.067 is measured over a strength range that a competent Reversi program would sit entirely above, and a wider range could show a relationship this one cannot see.

**A game is not a natural object.** A person put its regularity there deliberately. This is a positive control, in the same class as a published cell edge and with less excuse, since nobody has to go and measure a subtraction game. A positive control shows what a memoryless control cannot, and nothing larger is claimed for it here.

## 6.3 WHAT THE RESEARCH PAPER IS FOR, IF ONE THING

Not the three hundred and eighty three. They are cheap by design, and a domain that passes out its own answers is not hard to agree with.

The chapter worth carrying forward is the chapter where the instrument is confident, clears every floor by three to twenty nine times, produces a number with a clean mechanism behind it, and is still not reporting what its output is called. A memoryless control cannot find that, because a memoryless control has no structure to be confidently wrong about. It takes an object with real structure and a published answer saying the structure is not a period.

The test that catches it is free, and the instrument carries it.

# BIBLIOGRAPHY


[1] E. R. Berlekamp, J. H. Conway, R. K. Guy. *Winning Ways for your Mathematical Plays*. Academic Press. 1982.

[2] C. L. Bouton. *Nim, a game with a complete mathematical theory*. The Annals of Mathematics 3(1/4) 35, doi:10.2307/1967631. 1901.

[3] P. M. Grundy. *Mathematics and games*. Eureka 2, 6-8. 1939.

[4] M. Lothaire. *Algebraic Combinatorics on Words*. Cambridge University Press. 2002.

[5] D. Quigg. *Crystallography*. orior research papers, https://github.com/dstroy0/orior. 2026.

[6] D. Quigg. *Workbook*. orior research papers, https://github.com/dstroy0/orior. 2026.

[7] R. P. Sprague. *Über mathematische Kampfspiele*. Tôhoku Mathematical Journal 41, 438-444. 1935.

[8] W. A. Wythoff. *A modification of the game of Nim*. Nieuw Archief voor Wiskunde 7, 199-202. 1907.